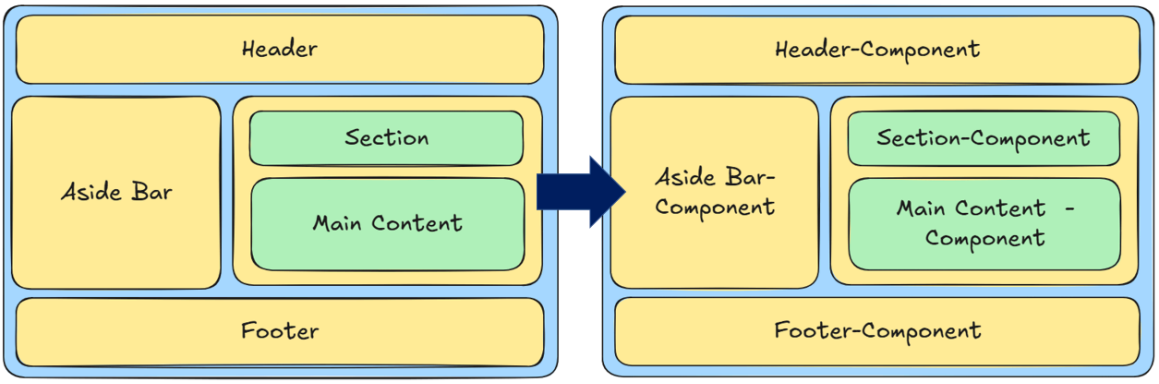
2. Characteristics of React JS

▼ 1. Component-Based Architecture

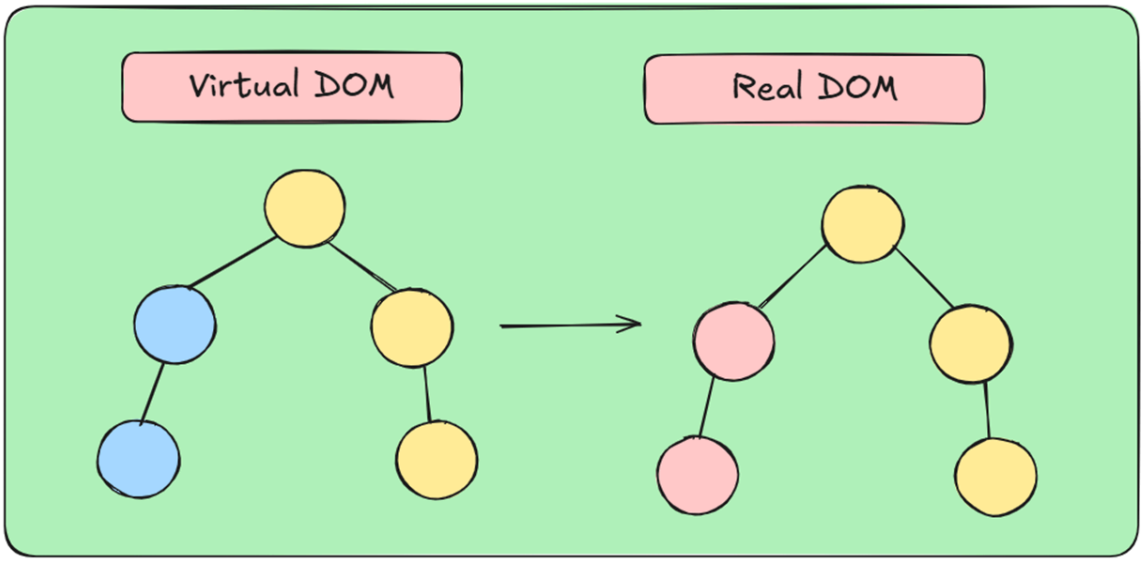
- React apps are made up of **small, reusable pieces** called **components**.
- Each component represents a **part of the UI** (like a button, header, form, etc.).
- Easy to manage and reuse across pages.
- Component architecture enables the developer to break down the large application into smaller components, which can be further break down into even smaller component.
- Breaking down the application into smaller component simplifies the application and make it more understandable and manageable.
- **For Example →**



▼ 2. Virtual DOM

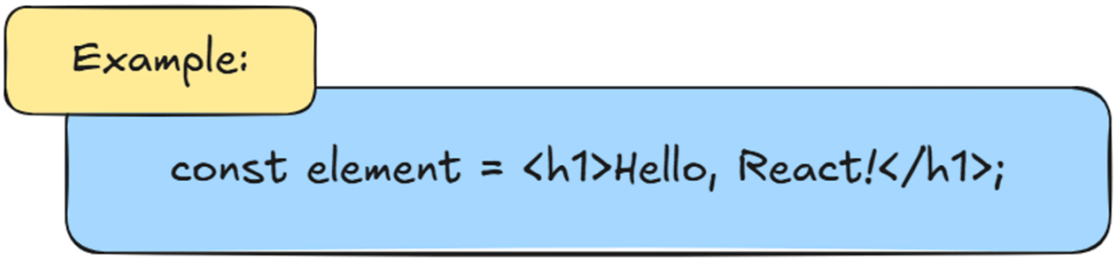
- React uses a **Virtual DOM.**
- Virtual DOM is a copy of the real DOM.
- Virtual DOM represents the real DOM of the current HTML document.
- It updates only the **changed parts**, not the whole webpage.
- This makes the app **faster and more efficient**.
- This improves the performance of the rendering of the HTML document.

- **For Example →**



▼ 3. JSX

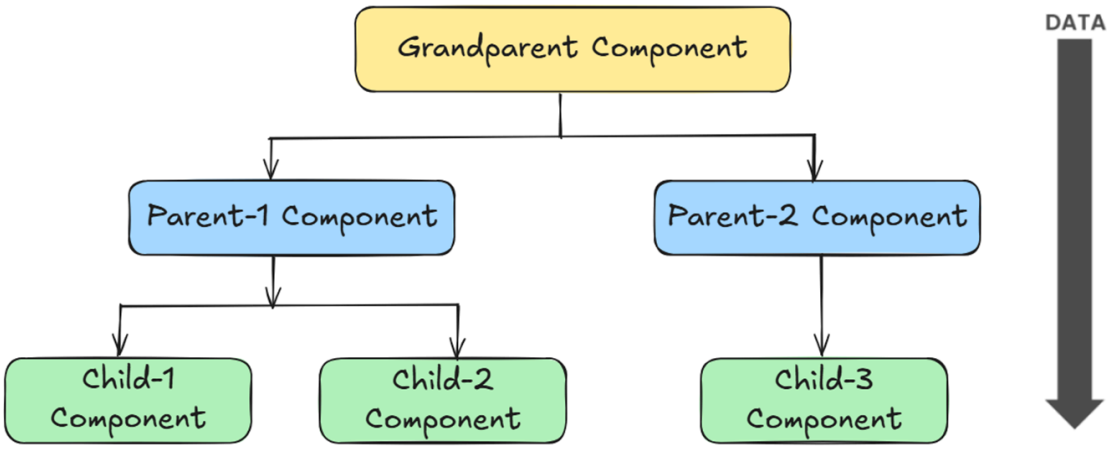
- JSX stands for **Javascript and XML**.
- JSX allows writing **HTML-like syntax inside JavaScript**.
- It looks like HTML but works like JavaScript.
- **For Example →**



▼ 4. One Way Data Binding (Unidirectional)

- One way data binding means data flows in a single direction.
- A component can pass the data to its child component only.

- The data cannot be passed by a component to its parent component in any situation.
- This will simplify the data handling and reduce the complexity.
- For Example: A **parent component** passes data to a **child component** via props.



▼ 5. Declarative Syntax

- You describe **what the UI should look like**, not how to do it step-by-step.
- React takes care of updating the UI when the data changes.
- Developers only describe **what** the UI should look like, and React handles the rest.
- **For Example →**

Imperative	Declarative
How to do things.	What to do.
Statements.	Expressions.
Defining variables and changing their values.	Evaluate result based on input.

▼ 6. Scalable

- React is great for **small or large apps**.
- You can build complex applications by combining simple components.
- Features like **components**, **Virtual DOM**, and **one-way data flow** help manage big apps easily and keep them fast.
- It works well even when your app gets more complex.

▼ 7. Flexible

- React gives you just the **basic tools** (like components and hooks).
- It doesn't force you to follow any specific rules or structure.
- You are **free to build your app your own way**, using the patterns you like.

▼ 8. Modular

- In React, you can write each component in its **own file**.
- You can then **group related components** into folders or modules.
- This makes your code **organized, reusable, and easy to manage**.